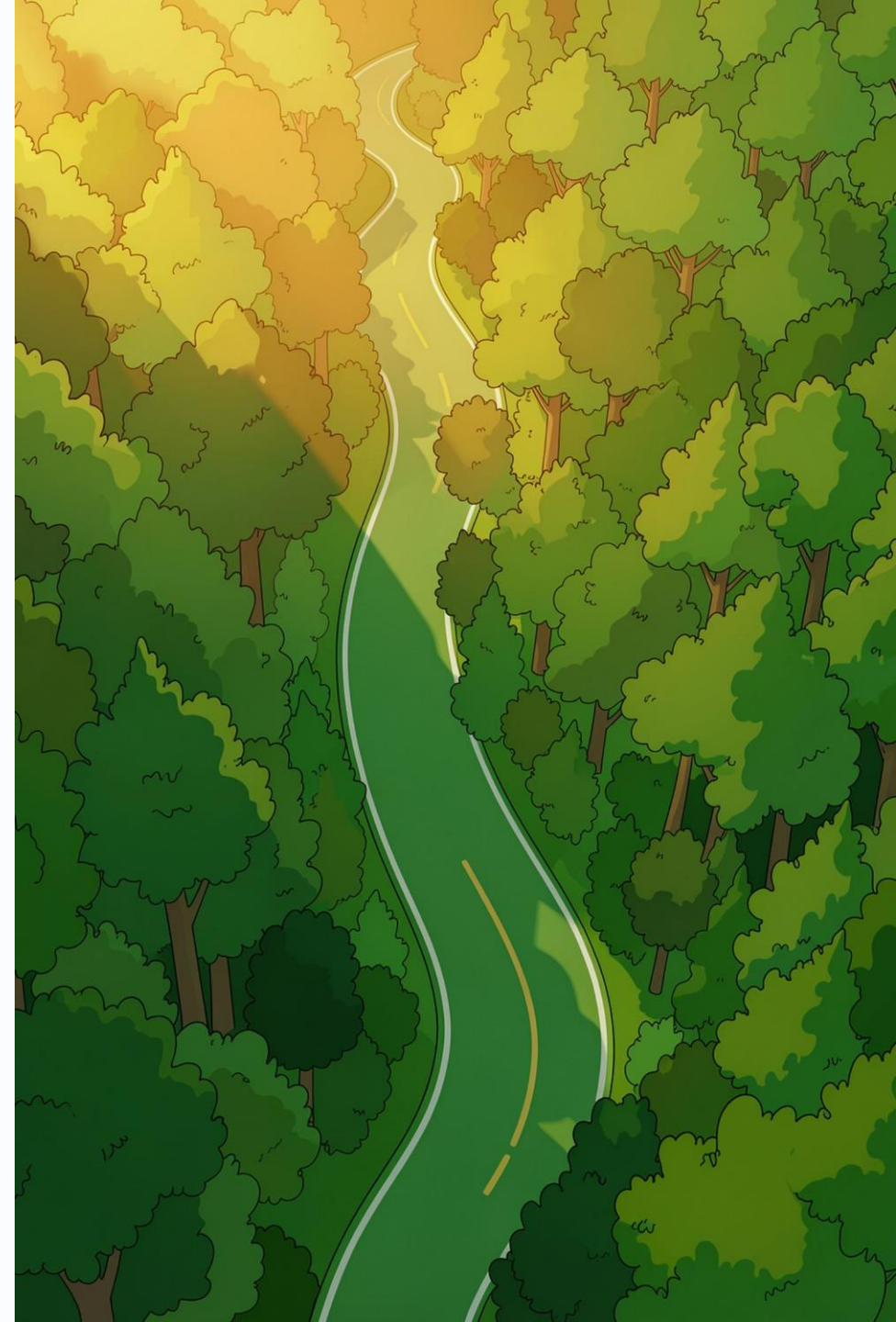


Eco-Route AI

Navigating a Greener Future

Smart route planning meets environmental responsibility — helping individuals and businesses make smarter, cleaner travel decisions.

Team	Roll no.
Raymond Maijo	19
Rebekah P Joy	20
Reyhan Ashik	21
Ria Benny	22
Rishikesh Sidharth	23
Riya Ramesh	24





The Challenge: Understanding Our Environmental Impact

Transportation is one of the largest contributors to global carbon emissions. Yet most of us travel every day without knowing the true environmental cost of our routes.

24% of Global CO₂

Transportation accounts for nearly a quarter of all energy-related carbon emissions worldwide

Hidden Costs

Most travelers have no visibility into the carbon footprint of their daily journeys

No Easy Alternatives

Finding greener routes requires time, data, and tools most people simply don't have

Build your trip

Choose a corridor location and the exact vehicle model year you want to model.

FROM



Aluva Railway Station



TO



Rajagiri School of Engineering



VEHICLE TYPE



4 Wheeler



2 Wheeler

VEHICLE MODEL



Maruti Suzuki Swift · Petrol



0.142 kg CO₂/km base factor · 0.050 g NO₂/km base factor

MODEL YEAR



2022 model year



Older model years receive a small age-adjustment in the classroom estimate.

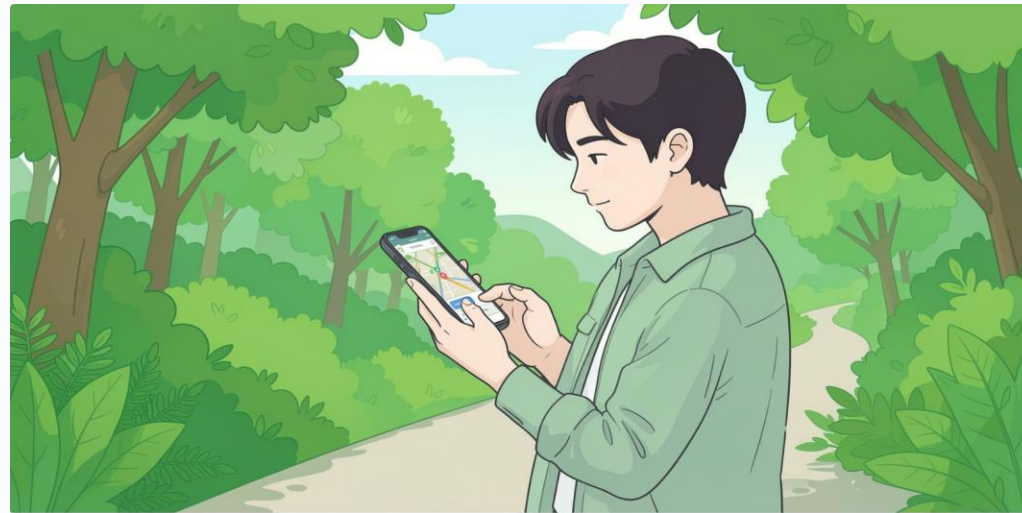


Current traffic

Light

30 km/h current speed · 30 km/h free-flow · road class FRC4

Updated 08:33 PM



Introducing Eco-Route AI

Eco-Route AI is an intelligent, AI-powered platform that transforms how you plan your journeys — combining real-time route optimization with personalized carbon footprint data to make sustainable travel effortless.

- AI-driven route recommendations
- Real-time carbon emission tracking
- Interactive map visualization
- Accessible for everyone, everywhere

MEET YOUR COMPANION

Personalized Carbon Footprint Calculation

Eco-Route AI doesn't just show you a route — it shows you the **environmental cost** of every journey.



Instant Emissions Data

Enter any origin and destination to receive an immediate, accurate carbon footprint estimate for your trip.



Vehicle-Aware Calculations

Input your vehicle type for tailored calculations that reflect your real-world emissions profile.

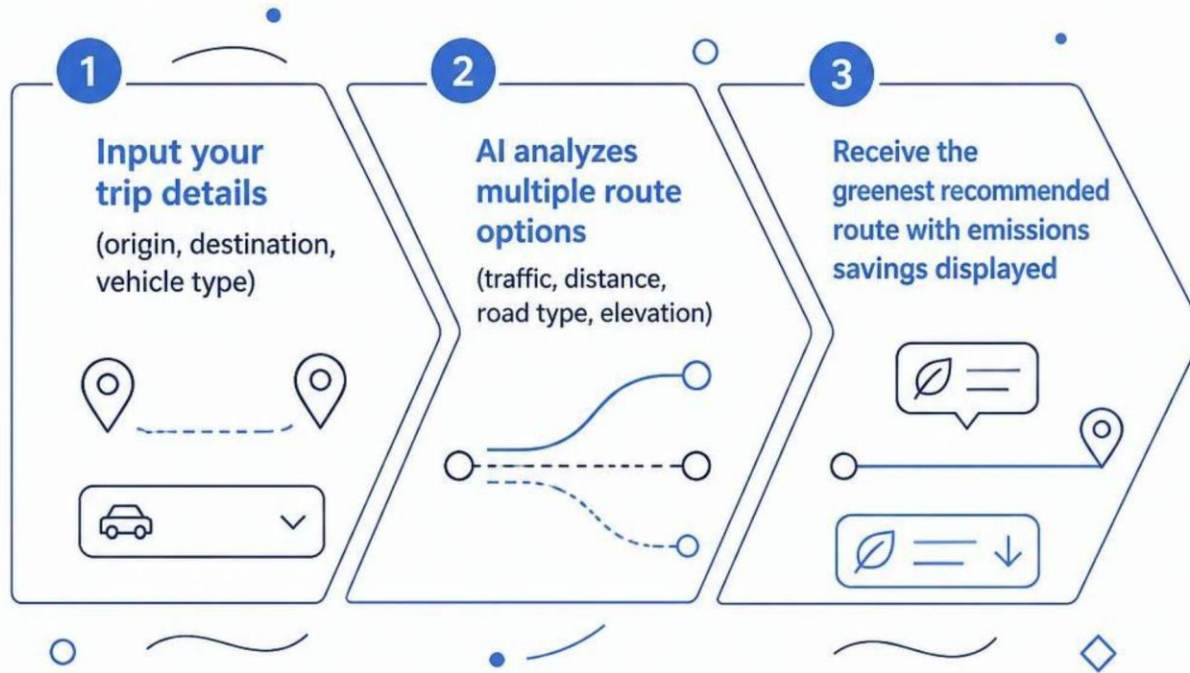


Mode Comparison

Compare emissions across driving, cycling, public transit, and walking to find your best option.



Optimized Eco-Friendly Route Planning



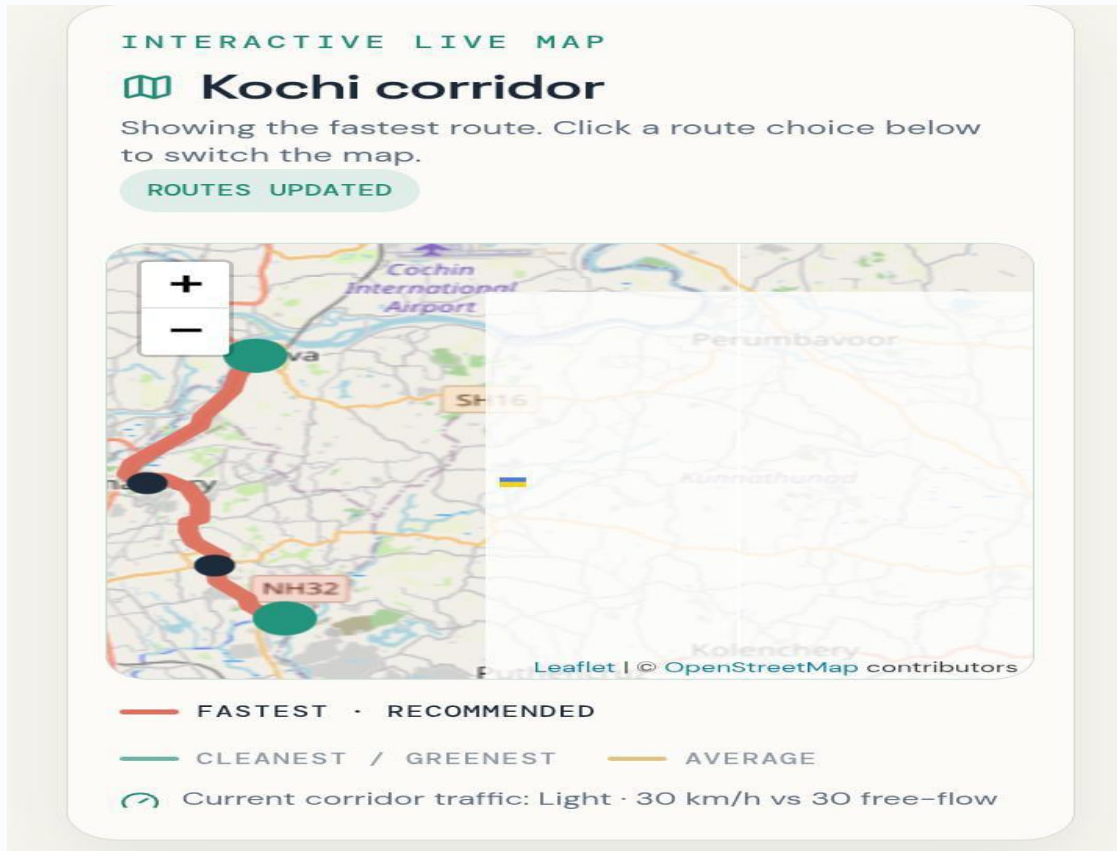
ROUTE CHOICE	TIME	DISTANCE	CO ₂	NO ₂	FIT
 Fastest route Best for the shortest travel time	19 min	16.5 km	2.39 kg	0.84 g	✓ Recommended
 Cleanest / greenest route Lower fuel burn and emissions	22 min	17.2 km	2.49 kg	0.88 g	● 82
 Average route Balanced distance and travel time	30 min	22.1 km	3.20 kg	1.13 g	● 0

Smarter Routes, Lower Emissions

Eco-Route AI evaluates multiple routing variables simultaneously — including traffic patterns, road gradients, stop frequency, and distance — to surface the most fuel-efficient path available.

- Every optimized route displays estimated fuel savings and CO₂ reduction compared to the standard fastest route.

Interactive Map Visualization & Data



See Your Impact Visually

Eco-Route AI's interactive map brings your travel data to life with intuitive visualizations that make environmental impact easy to understand at a glance.

Color-Coded Routes

Green-to-red gradients instantly communicate emission levels across route options

Historical Tracking

Review past trips and monitor your cumulative carbon savings over time

Data Export

Download trip reports for personal records or corporate sustainability reporting

🕒 30 min 📍 22.1 km ☁️ 3.20 kg CO₂ 🚗 1.13 g NO₂

Maruti Suzuki Swift

0/100 fit

OTHER WAYS TO TRAVEL

Compare bus, rail, and vehicle access

Transit is shown only when the first and last mile can be walked in 750 m or less; longer access legs use the selected vehicle.

PER PASSENGER

City bus



Shared road transport

Start 420 m walk · end 620 m walk

1.47 kg CO₂ 0.007 g NO₂

Auto-rickshaw



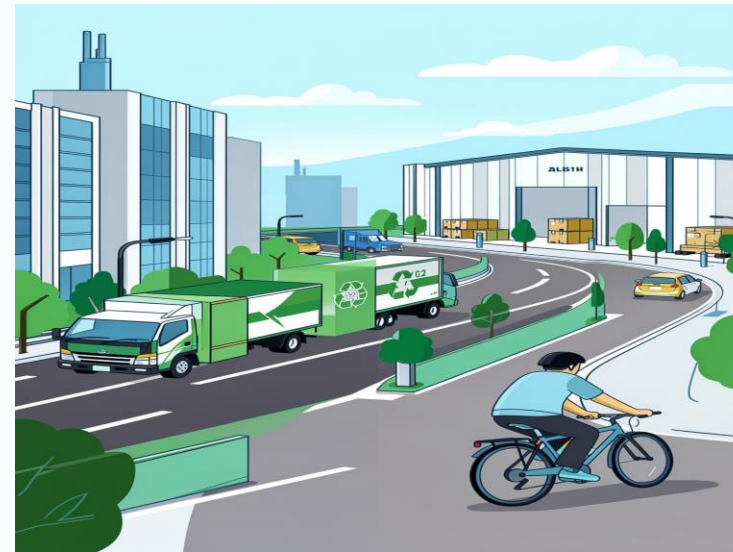
Short-distance shared ride

Direct road-based trip

1.98 kg CO₂ 0.010 g NO₂

ⓘ Bus and rail availability use the selected Kochi place and nearby access-point estimates. Distances over 750 m are explicitly assigned to the selected 4 Wheeler / 2 Wheeler instead of being treated as walking.

[View full route comparison >](#)



Real-World Applications



Daily Commutes

Individuals reduce personal emissions by choosing optimized routes for everyday travel — saving fuel and money.



Business Logistics

Fleet managers and delivery companies optimize multi-stop routes to cut operational costs and meet sustainability targets.



Urban Planning

Cities and municipalities leverage aggregate data to identify high-emission corridors and plan greener infrastructure.



DESIGNED FOR EVERYONE

Intuitive Design & Accessibility

Eco-Route AI is built with a **user-first philosophy** — powerful AI capabilities wrapped in a clean, approachable interface that anyone can use without a technical background.

→ Zero Learning Curve

Simple input fields and clear visual outputs make it usable from the first tap

→ Mobile-Ready

Fully responsive design works seamlessly on desktop, tablet, and mobile devices

→ Language Support

Designed for global accessibility with multi-language support

The Impact: Fostering Sustainable Habits

Eco-Route AI doesn't just plan routes — it **changes behavior** by making environmental impact visible, actionable, and rewarding.

24%

CO₂ Reduction Potential

Average emissions saved per optimized trip versus standard routing

1M+

Trips Tracked

Projected annual trip calculations as adoption scales globally

3x

Engagement Boost

Users who visualize their carbon data are 3x more likely to choose greener routes consistently

5

Transport Modes

Supported modes including car, EV, bus, cycling, and walking for comprehensive trip planning